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Short communication

Problematic smartphone use and mental health outcomes among Singapore residents: The health and lifestyle survey

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ABSTRACT

Background: The excessive use of smartphones and its association with adverse outcomes has been widely reported, with several studies showing an association between smartphone overuse, depression, anxiety, and sleep-related problems.

Methods: The study used data from the Health and Lifestyle Survey, a nationwide population survey. It examined the prevalence of Problematic Smartphone Use (PSU) and its association with mental health outcomes among Singapore residents aged 15–65 years.

Participants: Participants were assessed for PSU using the Smartphone Addiction Scale-Short Version, psychological distress with the Patient Health Questionnaire-9, and Generalised Anxiety Disorder −7 questionnaire, sleep problems using the Insomnia Severity Index, and positive mental health with Rapid Positive Mental Health Instrument.

Results: In all, 6509 participants completed the survey, giving a survey response rate of 73.2 %. The prevalence of PSU was 30.2 % in the population. Individuals with PSU were more likely to have symptoms of moderate or severe depression (OR: 3.2, 95 % CI: 2.4–4.4), anxiety (OR: 3.4, 95 % CI: 2.4–4.8), insomnia (OR: 3.4, 95 % CI: 2.8–4.2), and poorer positive mental health (β: −0.3, 95 % CI: −0.4 to −0.2).

Conclusions: The study is the first to examine PSU in a national sample of Singaporeans across a wide age range. It provides valuable insights into mental health comorbidities among those with PSU, which is useful for practitioners.

1. Introduction

The wide availability of smartphones globally and its ability to connect to the Internet from almost anywhere has drastically changed the lives of people across all walks of life by enabling easy access to information, services, entertainment, communication, and navigation. However, the excessive use of smartphones and its association with adverse outcomes is increasingly being reported. The consequences of smartphone overuse include the loss of productivity due to constant interruptions (Duke and Montag, 2017), difficulty disengaging from work at home (Derks et al., 2014; Schlachter et al., 2018), and less focus on quality family time. The risk of vehicular accidents has also increased because drivers get distracted by looking, holding, or talking on the phone (Atwood et al., 2018; Cao et al., 2022). Excessive smartphone use is also associated with adverse mental health outcomes. Several studies have shown an association between smartphone overuse, depression, anxiety (Alhassan et al., 2018; Elhai et al., 2017), and insufficient sleep duration or sleep quality (Lee et al., 2017; Nikolic et al., 2023).

In Singapore, the number of smartphone users reached about 5.57 million in 2022 (Lucas Romero, 2023. Smartphone users in Singapore 2019-2028. In *Statista* (September 12, 2023). Retrieved from https://www.statista.com/statistics/494598/smartphone-users-in-singapore/?). According to the Infocomm Media Development Authority (IMDA), 97 % of

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residents aged 18 and above own a smartphone (Infocomm Media Development Authority., 2023). Given the lack of local data and the high prevalence of smartphone ownership, the current study aimed to examine the prevalence of problematic smartphone use (PSU), its socio-demographic correlates, and its association with mental health outcomes in Singapore.

2. Methods

2.1. Study Design

Data for the study came from the Health and Lifestyle Survey - a nationwide population survey that primarily aimed to establish the prevalence of behavioral and substance addictions among Singapore residents aged 15–65 years. The study recruited 6509 participants using a disproportionate stratified sampling, i.e., those of Chinese, Malay, and Indian ethnicity were sampled in equal proportions (vs. Chinese 74.3 %, Malay 13.5 %, and Indian 9 % in the population). The survey was conducted over 15 months, from April 2021 to July 2022. The study has been described in detail in an earlier publication (Subramaniam et al., 2024).

2.2. Participants

The sample comprised Singapore residents (Singapore Citizens and Permanent Residents) aged between 15 and 65 years, randomly selected from an administrative database that comprises socio-demographic data and residential addresses of all Singapore residents. The sampling thus included all Singapore residents aged 15–65 years. Other inclusion criteria were as follows: belonging to any of the four ethnic groups in Singapore (Chinese, Malay, Indian, Others) and being literate in English, Chinese, Malay, or Tamil. Those who were unable to complete the survey due to severe physical or mental health conditions and those who were institutionalised for the entire study period were excluded.

2.3. Sample size

The study's sample size was initially determined through statistical power calculations for binary proportions, aiming to achieve a margin of error of 0.05 for both the overall population and sub-groups. We set a statistical power of 0.80, with a Type 1 error at $\alpha = 0.05$. Considering oversampling by age and ethnicity, the design effect was calculated to be 1.765. As the main aim of the study was to estimate the prevalence of drug consumption in the population, and given that there was no local data that could be used as a reference, we generated two power calculations based on an approximation of prevalence rates between 1 % and 2 %, respectively. In each calculation, we assumed specific, realistic sample sizes (e.g., n=5500, 6000, 6500) and computed the margin of error for key quantities of interest. Using this approach, we determined that a sample size of 6500 would provide sufficient precision to estimate 1–2 % of drug consumption (Subramaniam et al., 2024). Since the current study was focused on PSU, we also generated a power calculation based on an approximation of the prevalence rate of PSU of 38.5 % in the Hong Kong population as a reference. We found that a sample size of 6500 would provide high precision with a relative standard error of 2.1 % and a margin of error of 1.6 % for the overall prevalence estimate of PSU in our sample.

2.4. Survey methodology

Participants were sent an invitation letter and approached by trained lay interviewers using door-knocks. A verbal consent was sought in person from all the participants, following which the interviewer passed the tablet with the programmed questionnaire to the participant who completed them. After the imposition of COVID-19-related restrictions, participants were given a unique link to access the questionnaire online for completing the survey after administration of the verbal consent by the interviewers. In all, 414 participants completed the survey on a tablet and 6095 completed it using the online questionnaire.

2.5. Questionnaires

The Smartphone Addiction Scale-Short Version (SAS-SV) (Kwon et al., 2013) was used to assess PSU. A gender-specific cut-off of ≥ 31 and ≥ 33 for male and female participants was used, respectively.

The Patient Health Questionnaire - 9 (PHQ–9) (Kroenke et al., 2001) was used to screen for depression over the past two weeks. The General Anxiety Disorder −7 (GAD-7) (Spitzer et al., 2006) was used to identify probable cases of anxiety over the preceding two-week period. The Insomnia Severity Index (ISI) was used to assess the nature, severity, and impact of insomnia (Bastien et al., 2001; Morin et al., 2011) in the "last month." A cut-off of 10 was used to screen for depression, anxiety, and insomnia.

The Rapid-Positive Mental Health Instrument (R-PMHI) is a six-item unidimensional measure of positive mental health (PMH) (Vaingankar et al., 2020a). The total PMH score is obtained by averaging the item scores. A higher score indicates better positive mental health.

Information on the sociodemographic profile of the respondents, including age, gender, ethnicity, living circumstances, marital status, and education level, were also collected.

The questionnaires were translated into Chinese, Malay, and Tamil using standard techniques. Language and domain experts identified hard-to-understand words and phrases that were subsequently modified by the team. Questionnaires were tested cognitively among 48 Singapore residents, ensuring appropriate language use. Pilot testing was conducted in all languages (428 English, 74 Malay, 74 Tamil, 74 Chinese), and participants were asked to highlight any challenges they encountered. No problems with the words or phrases were identified at the pilot testing phase.

2.6. Ethical considerations

All study-related procedures were performed in compliance with the relevant laws and institutional guidelines and were approved by the National Healthcare Group Domain Specific Review Board (NHG DSRB). The DSRB also provided ethical oversight of the study (2019/00775 Approved on 31 October 2019).

2.7. Statistical analysis

Confirmatory factor analyses (CFA) were performed to determine the model fit of the one-factor model as proposed by the developers of SAS-SV (Kwon et al., 2013) and the three-factor model as presented by a study in Hong Kong (Cheung et al., 2019). The three factors comprise problems (3 items), dependency (5 items), and time spent (2 items).

Sociodemographic correlates of PSU (age group, gender, ethnicity, education, marital status, employment and income) and associations with mental disorders were examined using logistic regression models. A linear regression model was used to examine the association of PSU (independent variable) with positive mental health (dependent variable). All analysis was weighted to account for the disproportionate stratified sampling design, non-response, and post-stratification by age, gender, and ethnicity. Standard errors were generated using Taylor's linearization. The analysis was performed using IBM SPSS version 23 and STATA M/P 18.0, with two-sided tests at a significance level of 5 %.

3. Results

3.1. Sociodemographic characteristics of the sample

In all, 6509 participants completed the survey, giving a survey response rate of 73.2 %. A total of 5536 participants were included in the

current analysis. Those with missing data, i.e., those who did not provide replies to all the items, were excluded from the analyses. The sociodemographic characteristics of the sample are shown in Table 1.

3.2. CFA of Smartphone Addiction Scale

The CFA results showed that the one-factor model had a poor model fit. On the other hand, the three-factor model had an acceptable model fit. A second-order model was considered more appropriate as the correlations between the three-factor model were high (0.72–0.86). The second-order three-factor model generated showed an acceptable model fit (CFI: 0.97, TLI: 0.96, SRMR: 0.03, RMSEA: 0.07) (Fig. 1). The internal consistency of PSU was good, as Cronbach's alpha was 0.89. Standardized factor loadings for the three factors were high, with values ranging from 0.71 to 0.85 (Table 2).

Table 1
Characteristics of the sample (n=5536).

	Weighted %	Unweighted n
Age Groups (years)		
15 – 34	37.8 %	2146
35 – 49	31.5 %	1939
50 – 65	30.7 %	1451
Gender		
Female	50.7 %	2839
Male	49.3 %	2697
Ethnicity		
Chinese	74.4 %	1873
Malay	13.1 %	1769
Indian	9.1 %	1637
Others	3.4 %	257
Education		
No formal education/Primary school	6.4 %	382
Secondary School	22.2 %	1244
Pre-tertiary education[#]	32.7 %	1983
Degree/Professional Qualification and Post Graduate and above	38.7 %	1860
Others	0.01 %	2
Marital Status		
Married	56.6 %	3220
Single	37.8 %	1949
Separated / Widowed / Divorced	5.6 %	320
Employment		
Employed/Self-employed	76.0 %	4059
Economically inactive/Students	20.5 %	1202
Unemployed/Temporarily laid off	3.4 %	193
Others	0.04 %	1
Personal Monthly Income (SGD)		
No income/Below 2000	41.3 %	2037
2000–3999	24.3 %	1242
4000–6999	19.1 %	740
7000 and above	15.3 %	549
Moderate or Severe Depression (PHQ-9)		
No	93.7 %	5116
Yes	6.3 %	418
Moderate or Severe Anxiety (GAD-7)		
No	94.7 %	5185
Yes	5.3 %	349
Moderate or Severe Insomnia (ISI)		
No	76.0 %	4114
Yes	24.0 %	1422
Doctor-diagnosed depression		
No	97.6 %	5264
Yes	2.5 %	140
Doctor-diagnosed anxiety		
No	97.6 %	5252
Yes	2.4 %	129

Missing data: Education (n=65), Marital status (n=47), employment (n=81), personal income (n=968), moderate or severe depression (n=2), moderate or severe anxiety (n=2), self-report of doctor-diagnosed depression (n=132), self-report of doctor-diagnosed anxiety (n=155)

[#] Pre-tertiary education consists of Vocational Institute/Pre-University/Junior College/Diploma

3.3. Prevalence of problematic smartphone use and its association with sociodemographic factors

Using cut-off values of ≥ 31 and ≥ 33 for male and female participants, respectively, the prevalence of PSU was 30.2 % (n=1578). The sociodemographic factors associated with smartphone use are shown in Table 3.

3.4. Association with mental disorders and positive mental health

Individuals with problematic smartphone usage were more likely to have symptoms of moderate or severe depression, symptoms of moderate or severe anxiety, moderate or severe insomnia, and lower positive mental health scores (Table 4). However, the R-square for the linear regression model with positive mental health as the outcome was 0.046, which is small. The effect size for this association was also small.

4. Discussion

A second-order three-factor model for SAS-SV provided a good fit for our data. The prevalence of PSU was 30.2 % (n=1578) among Singapore residents aged 15–65 years. It was lower than the weighted prevalence of PSU of 38.5 % in a population-based telephone survey in Hong Kong among adults aged 18 years and above (Luk et al., 2018).

Those belonging to the youngest age group, i.e., those aged 15–34, were most likely to experience PSU. Other studies have reported similar findings (Alhassan et al., 2018; Luk et al., 2018; de-Sola et al., 2017). The dependence of younger respondents on smartphones for socialisation purposes, coupled with lower self-control (Billieux, 2012) and higher social stress and anxiety (de-Sola et al., 2016), may predispose them to PSU. In addition, as this study was conducted during the COVID-19 pandemic, it is possible that many young people may have spent additional time communicating/ chatting with their peers on their smartphones.

PSU in the current study was associated with psychological distress, i.e., symptoms of moderate or severe depression and anxiety. Systematic reviews have shown that bivariate, unadjusted correlations between the severity of depression and anxiety and PSU had medium to small effect sizes (Elhai et al., 2019; Elhai et al., 2017). According to the Compensatory Internet Use Theory (CIUT) (Kardefelt-Winther, 2014), people attempt to reduce negative emotions by engaging in excessive internet use. As the smartphone is constantly accessible and available for use, those in distress may turn to using it to feel better. However, in a study of college students, high mobile phone use was associated with prolonged stress, sleep difficulties, and depressive symptoms at one-year follow-up (Thomée et al., 2011). Causal models suggest that smartphone use at night could impair sleep and influence stress and depression (Lemola et al., 2015; Andersen et al., 2023).

In our study, PSU was associated with moderate and severe insomnia. Studies examining the impact of smartphones on sleep have concluded that smartphone use in bed before sleep was associated with less sleep time, difficulty in sleeping, and excessive daytime sleepiness (Schweizer et al., 2017; Lemola et al., 2015; Munezawa et al., 2011). The reasons for poor sleep include sleep displacement by smartphone use (Correa-Iriarte et al., 2023), delayed onset of melatonin production by the electromagnetic radiation emitted by mobile phones before sleep (Wood et al., 2006), and cognitive arousal leading to insomnia (Kheirinejad et al., 2023).

Given the association with symptoms of depression, anxiety, and insomnia, it is not surprising that PSU was associated with lower positive mental health (Vaingankar et al., 2020b). Several studies have found a similar association between PSU and poor subjective well-being (Horwood and Anglim, 2019; Li et al., 2015; Studer et al., 2022).

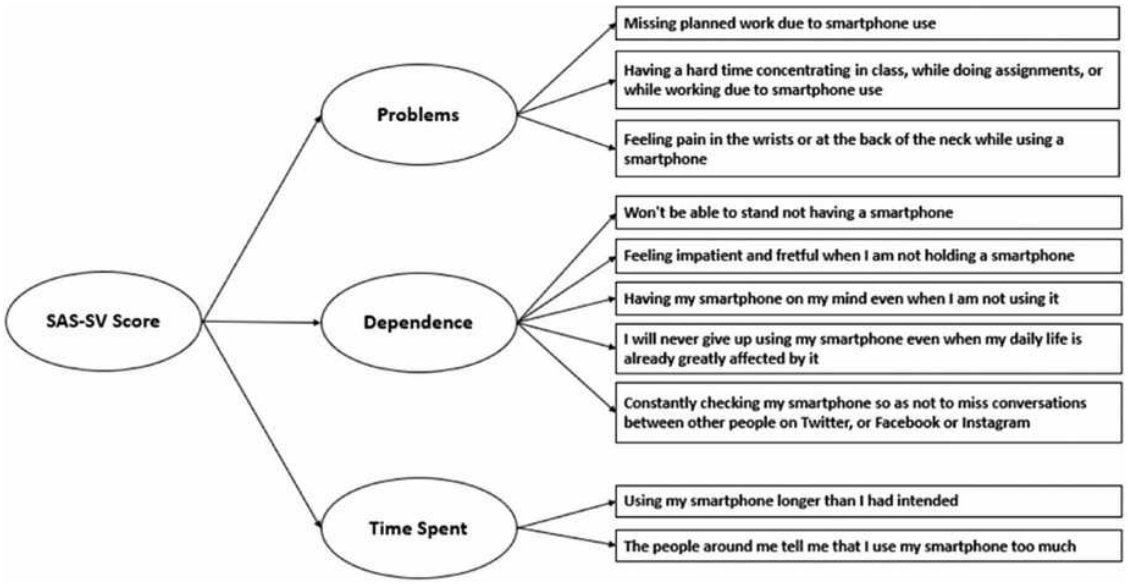


Fig. 1. Diagram to illustrate the second-order three-factor model of the Smartphone Addiction Scale-Short Version.

Table 2
Factor loadings of the three-factor model of the Smartphone Addiction Scale-Short Version.

Factor	Item	Standardized factor loading
Problems	Missing planned work due to smartphone use	0.79
	Having a hard time concentrating in class, while doing assignments, or while working due to smartphone use	0.85
	Feeling pain in the wrists or at the back of the neck while using a smartphone	0.74
Dependence	Won't be able to stand not having a smartphone	0.71
	Feeling impatient and fretful when I am not holding a smartphone	0.84
	Having my smartphone on my mind even when I am not using it	0.84
	I will never give up using my smartphone even when my daily life is already greatly affected by it	0.79
	Constantly checking my smartphone so as not to miss conversations between other people on Twitter, or Facebook or Instagram	0.75
Time spent	Using my smartphone longer than I had intended	0.80
	The people around me tell me that I use my smartphone too much	0.78

Table 3
Sociodemographic Correlates of Problematic Smartphone Use.

	PSU	
	Adjusted OR[§] (95 % CI)	p-value
Age Groups		
15–34 (Reference)		
35–49	0.6 (0.5–0.7)	< **0.001**
50–65	0.3 (0.2–0.4)	< **0.001**
Gender		
Female (Reference)		
Male	1.0 (0.9–1.2)	0.838
Ethnicity		
Chinese (Reference)		
Malay	0.9 (0.7–1.1)	0.156
Indian	0.6 (0.5–0.7)	< **0.001**
Others	0.7 (0.5–1.0)	0.059
Education		
Degree/Professional Qualification and Post Graduate and above (Reference)		
No formal education/Primary school	0.9 (0.5–1.5)	0.627
Secondary School	1.3 (1.0–1.8)	0.066
Pre-tertiary education[#]	1.3 (1.02–1.6)	**0.033**
Marital Status		
Married (Reference)		
Single	1.1 (0.8–1.4)	0.563
Separated / Widowed / Divorced	1.6 (1.05–2.4)	**0.029**
Employment		
Employed/Self-employed (Reference)		
Economically inactive[±]	1.0 (0.8–1.4)	0.827
Unemployed/Temporarily laid off	0.9 (0.5–1.5)	0.608
Personal Monthly Income (SGD)		
No income/Below 2000 (Reference)		
2000–3999	1.2 (0.9–1.6)	0.251
4000–6999	1.3 (0.9–1.8)	0.174
7000 and above	1.5 (1.0–2.1)	0.055

[#] Pre-tertiary education consists of Vocational Institute/ Pre-University/Junior College/and Diploma
[±]Economically inactive: Students, Homemakers and Retired
[§] Model adjusted for age group, gender, ethnicity, education, marital status, employment and income.

4.1. Strengths and Limitations

The strengths of our study include a community sample comprising adolescents and adults who were recruited by door-knocks. The high response rate (73.2 %) ensures the generalisability of our results to the population of Singapore. Translating the questionnaires into all four major languages spoken in Singapore ensured inclusivity. The self-administered mode of the surveys and assurance of data confidentiality reduced the risk of social desirability bias to a large extent. On the other hand, the study has some limitations. While the SAS-SV is a widely used assessment tool to identify PSU, we could not confirm the diagnosis by a clinical assessment. Another limitation is the small R-square value and weak effect size found in the association between PSU and positive mental health. However, as positive mental health is an emerging field in Singapore, this finding can still inform relevant organizations on strategies to promote positive mental health in the country. Lastly, the cross-sectional nature of our study limits us from drawing causal

Table 4
Association of Mental Health Conditions and Positive Mental Health with Problematic Smartphone Use.

	PSU	
Binary Outcomes	OR (95 % CI)	p-value
Moderate or Severe Depression	3.2 (2.4–4.4)	**< 0.001**
Moderate or Severe Anxiety	3.4 (2.4–4.8)	**< 0.001**
Moderate or Severe Insomnia	3.4 (2.8–4.2)	**< 0.001**
Continuous Outcome	β (95 % CI)	p-value
Positive Mental Health	-0.3 (-0.4 to -0.2)	**< 0.001**

All regression models adjusted for age group, gender, ethnicity, education, marital status, employment and income.

conclusions between PSU and mental disorders.

In conclusion, the prevalence of PSU in Singapore was 30 %, translating into about 1 in 3 people in the general population having PSU. The association between PSU, anxiety, depression, and insomnia is concerning. Longitudinal studies are needed to understand the association between PSU and adverse health outcomes further.

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CRediT authorship contribution statement

Siow Ann Chong: Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Wai Leng Chow:** Writing – review & editing, Resources, Conceptualization. **Yen Sin Koh:** Writing – review & editing, Formal analysis. **Mythily Subramaniam:** Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Ellaisha Samari:** Writing – review & editing, Supervision, Project administration, Methodology. **Rajeswari Sambasivam:** Writing – review & editing, Supervision, Project administration, Methodology. **Anitha Jeyagurunathan:** Writing – review & editing, Project administration. **Edimansyah Abdin:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Yunjue Zhang:** Writing – review & editing, Supervision, Project administration. **Bernard Tan:** Writing – review & editing, Formal analysis, Data curation. **Stefan Ma:** Writing – review & editing, Software, Data curation.

Declaration of Competing Interest

The authors have no Conflicts of Interest to declare.

References

Alhassan, A.A., Alqadhib, E.M., Taha, N.W., Alahmari, R.A., Salam, M., Almutairi, A.F., 2018. The relationship between addiction to smartphone usage and depression among adults: a cross sectional study. BMC Psychiatry 18 (1), 148. https://doi.org/10.1186/s12888-018-1745-4.

Andersen, T.O., Sejling, C., Jensen, A.K., Drews, H.J., Ritz, B., Varga, T.V., Rod, N.H., 2023. Nighttime smartphone use, sleep quality, and mental health: investigating a complex relationship. Sleep 46 (12), zsad256. https://doi.org/10.1093/sleep/zsad256.

Atwood, J., Guo, F., Fitch, G., Dingus, T.A., 2018. The driver-level crash risk associated with daily cellphone use and cellphone use while driving. Accid. ; Anal. Prev. 119, 149–154.

Bastien, C.H., Vallières, A., Morin, C.M., 2001. Validation of the insomnia severity index as an outcome measure for insomnia research. Sleep. Med. 2, 297–307.

Billieux, J., 2012. Problematic use of the mobile phone: a literature review and a pathways model. Curr. Psychiatry Rev. 8 (4), 299–307.

Cao, X., Cheng, Y., Xu, C., Hou, Y., Yang, H., Li, S., Gao, Y., Jia, P., WangY, 2022. Risk of accidents or chronic disorders from improper use of mobile phones: a systematic review and meta-analysis. J. Med. Internet Res. 24 (1), e21313 https://doi.org/10.2196/21313.

Cheung, T., Lee, R.L.T., Tse, A.C.Y., Do, C.W., So, B.C.L., Szeto, G.P.Y., Lee, P.H., 2019. Psychometric properties and demographic correlates of the smartphone addiction scale-short version among Chinese children and adolescents in Hong Kong. Cyber, Behav. Soc. Netwworking 22 (11), 714–723.

Correa-Iriarte, S., Hidalgo-Fuentes, S., Martí-Vilar, M., 2023. Relationship between problematic smartphone use, sleep quality and bedtime procrastination: a mediation analysis. Behav. Sci. (Basel) 13 (10), 839.

Derks, D., Van Mierlo, H., Schmitz, E.B., 2014. A diary study on work-related smartphone use, psychological detachment and exhaustion: examining the role of the perceived segmentation norm. J. Occup. Health Psychol. 19 (1), 74.

de-Sola, G.J., Rodríguez, D.F.F., Rubio, G., 2016. Cell-phone addiction: a review. Front. Psychiatry 7, 175. https://doi.org/10.3389/fpsyt.2016.00175.

de-Sola, J., Talledo, H., de Fonseca, F.R., Rubio, G., 2017. Prevalence of problematic cell phone use in an adult population in Spain as assessed by the Mobile Phone Problem Use Scale (MPPUS). PLoS One 12 (8), e0181184. https://doi.org/10.1371/journal.pone.0181184.

Duke, É., Montag, C., 2017. Smartphone addiction, daily interruptions and self-reported productivity. Addict. Behav. Rep. 6, 90–95. https://doi.org/10.1016/j.abrep.2017.07.002.

Elhai, J.D., Dvorak, R.D., Levine, J.C., Hall, B.J., 2017. Problematic smartphone use: a conceptual overview and systematic review of relations with anxiety and depression psychopathology. J. Affect. Disord. 207, 251–259. https://doi.org/10.1016/j.jad.2016.08.030.

Elhai, J.D., Levine, J.C., Hall, B.J., 2019. The relationship between anxiety symptom severity and problematic smartphone use: a review of the literature and conceptual frameworks. J. Anxiety Disord. 62, 45–52. https://doi.org/10.1016/j.janxdis.2018.11.005.

Horwood, S., Anglim, J., 2019. Problematic smartphone usage and subjective and psychological well-being. Comput. Hum. Behav. 97, 44–50. https://doi.org/10.1016/j.chb.2019.02.028.

Infocomm Media Development Authority. (2023). Digital Society. ⟨https://www.imda.gov.sg/about-imda/research-and-statistics/digital-society⟩.

Kardefelt-Winther, D., 2014. A conceptual and methodological critique of internet addiction research: towards a model of compensatory internet use. Comput. Hum. Behav. 31, 351–354. https://doi.org/10.1016/j.chb.2013.10.059.

Kheirinejad, S., Visuri, A., Ferreira, D., Hosio, S., 2023. Leave your smartphone out of bed: quantitative analysis of smartphone use effect on sleep quality. Pers. Ubiquitous Comput. 27, 447–466. https://doi.org/10.1007/s00779-022-01694-w.

Kroenke, K., Spitzer, R., Williams, J., 2001. The PHQ-9. J. Gen. Intern. Med. 16 (9), 606–613.

Kwon, M., Kim, D., Cho, H., Yang, S., 2013. The smartphone addiction scale: development and validation of a short version for adolescents. Plos ONE 8 (12), e83558. https://doi.org/10.1371/journal.pone.0083558.

Lee, J.E., Jang, S.I., Ju, Y.J., Kim, W., Lee, H.J., Park, E.C., 2017. Relationship between mobile phone addiction and the incidence of poor and short sleep among Korean adolescents: a longitudinal study of the korean children & youth panel survey. J. Korean Med. Sci. 32 (7), 1166–1172. https://doi.org/10.3346/jkms.2017.32.7.1166.

Lemola, S., Perkinson-Gloor, N., Brand, S., Dewald-Kaufmann, J.F., Grob, A., 2015. Adolescents' electronic media use at night, sleep disturbance, and depressive symptoms in the smartphone age. J. Youth Adolesc. 44, 405–418. https://doi.org/10.1007/s10964-014-0176-x.

Li, J., Lepp, A., Barkley, J.E., 2015. Locus of control and cell phone use: Implications for sleep quality, academic performance, and subjective well-being. Comput. Hum. Behav. 52, 450–457.

Lucas Romero. Smartphone users in Singapore2019-2028. In Statista (September 12, 2023). Retrieved from https://www.statista.com/statistics/494598/smartphone-users-in-singapore/?_gl=1*z6z6a3*_ga*MjAwMDczNzU3NS4xNzE5MTkxNjgy*_ga_NY8L5SJPMB*MTcxOTE5MjM1MS4xLjEuMTcxOTE5MjM4NS4wLjAuMA.

Luk, T.T., Wang, M.P., Shen, C., Wan, A., Chau, P.H., Oliffe, J., Viswanath, K., Chan, S.S.-C., Lam, T.H., 2018. Short version of the Smartphone Addiction Scale in Chinese adults: psychometric properties, sociodemographic, and health behavioral correlates. J. Behav. Addict. 7 (4), 1157–1165. https://doi.org/10.1556/2006.7.2018.105.

Morin, C.M., Belleville, G., Bélanger, L., Ivers, H., 2011. The Insomnia Severity Index: psychometric indicators to detect insomnia cases and evaluate treatment response. Sleep 34, 601–608.

Munezawa, T., Kaneita, Y., Osaki, Y., Kanda, H., Minowa, M., Suzuki, K., Higuchi, S., Mori, J., Yamamoto, R., Ohida, T., 2011. The association between use of mobile phones after lights out and sleep disturbances among Japanese adolescents: a nationwide cross-sectional survey. Sleep 34, 1013–1020. https://doi.org/10.5665/SLEEP.1152.

Nikolic, A., Bukurov, B., Kocic, I., Vukovic, M., Ladjevic, N., Vrhovac, M., Pavlovic, Z., Grujicic, J., Sipetic, S., 2023. Smartphone addiction, sleep quality, depression, anxiety, and stress among medical students. Front. Public Health 11, 1252371. https://doi.org/10.3389/fpubh.2023.1252371.

Schlachter, S., McDowall, A., Cropley, M., Inceoglu, I., 2018. Voluntary work-related technology use during non-work time: a narrative synthesis of empirical research and research agenda. Int. J. Manag. Rev. 20 (4), 825–846.

Schweizer, A., Berchtold, A., Barrense-Dias, Y., Akre, C., Suris, J.C., 2017. Adolescents with a smartphone sleep less than their peers. Eur. J. Pediatr. 176 (1), 131–136.

Spitzer, R.L., Kroenkem, K., Williamsm, J.B., Löwem, B., 2006. A brief measure for assessing generalized anxiety disorder: the GAD-7. Arch. Intern. Med. 166, 1092–1097.

Studer, J., Marmet, S., Wicki, M., Khazaal, Y., Gmel, G., 2022. Associations between smartphone use and mental health and well-being among young Swiss men. J. Psychiatr. Res. 156, 602–610.

Subramaniam, M., Koh, Y.S., Sambasivam, R., Abdin, E., Asharani, P.V., Vaningankar, J., Chua, B.Y., Chua, H.C., Lee, C., Chow, W.L., Ma, S., Chong, S.A., 2024. Prevalence of consumption of illicit drugs and associated factors from a nationwide epidemiological survey: the Singapore Health and Lifestyle Survey. Ann. Acad. Med Singap. 53 https://doi.org/10.47102/annals-acadmedsg.2023347.

Thomée, S., Härenstam, A., Hagberg, M., 2011. Mobile phone use and stress, sleep disturbances, and symptoms of depression among young adults–a prospective cohort study. BMC Public Health 11, 66. https://doi.org/10.1186/1471-2458-11-66.

Vaingankar, J.A., Abdin, E., van Dam, R.M., Chong, S.A., Tan, L.W.L., Sambasivam, R., Seow, E., Chua, B.Y., Wee, H.L., Lim, W.Y., Subramaniam, M., 2020a. Development and validation of the Rapid Positive Mental Health Instrument (R-PMHI) for measuring mental health outcomes in the population. BMC Public Health 20 (1), 471. https://doi.org/10.1186/s12889-020-08569-w.

Vaingankar, J.A., Chong, S.A., Abdin, E., Siva Kumar, F.D., Chua, B.Y., Sambasivam, R., Shafie, S., Jeyagurunathan, A., Seow, E., Subramaniam, M., 2020b. Understanding the relationships between mental disorders, self-reported health outcomes and positive mental health: findings from a national survey. Health Qual. Life Outcomes 18, 55. https://doi.org/10.1186/s12955-020-01308-0.

Wood, A.W., Loughran, S.P., Stough, C., 2006. Does evening exposure to mobile phone radiation affect subsequent melatonin production? Int. J. Radiat. Biol. 82, 69–76.